

MANOJ ADHIKARI

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PROFESSIONAL SUMMARY

Data Analyst with 5 years of advanced data analytics and engineering experience in Insurance, Ecommerce, and EdTech domain. Specializes in building robust data pipelines, data warehousing, semantic modeling, uncovering patterns, conducting statistical analysis, and developing predictive models to support data-driven decision making. Demonstrates expertise in data governance, security best practices, and Agile methodologies. Actively seeking opportunities to grow in fast-paced, dynamic, initiative-driven teams where data is at the core of strategic impact.

TECHNICAL SKILLS

Programming	Python, SQL, R
Database	SQL (MySQL, PostgreSQL), NoSQL (MongoDB, Elasticsearch), Amazon Redshift
BI tools & Visualization	PowerBI, Tableau, Amazon QuickSight, Looker, Excel, Streamlit
Cloud & Tools	Apache Airflow, Amazon SageMaker, Amazon S3, AWS Lambda, AWS Glue, Azure AI Studio, PySpark, Spark Streaming, Git, Docker, Pandas, NumPy, CuPy, REST APIs
AI/ML & NLP	GenAI, LLMs, TensorFlow, PyTorch, Scikit-learn, Transformers, RAG, SpaCy, Keras
Soft Skills:	Leadership, Stakeholder Engagement, Collaboration, Problem-Solving

PROFESSIONAL EXPERIENCE

R&D Engineer- AI & Analytics, East Tennessee State University Sep 2023 – April 2025 | Tennessee

- Analyzed **50K+** user queries from a university Q&A platform to identify recurring campus infrastructure issues, leading to an **18%** improvement in satisfaction post-survey.
- Build **incremental** and **interactive** dashboard and reports to communicate insights with university administration.
- Developed ETL for unstructured web text data, enabling incremental scraping, cleaning, and transformation with reliable scheduling and fault tolerance.
- Perform sentiment analysis on course feedback to segment and monitor sentiment trajectories of student.
- Developed interactive dashboards to support university administration, improving decision-making processes.
- Ensured **data governance and security** by implementing role-based access controls and maintaining compliance with university data privacy standards.
- Created detailed documentation for data collection, transformation, and analysis workflows and dashboard usage, ensuring clarity for non-technical users.

Data Analyst-II (Insurance), Renegade Insurance Feb 2022 - July 2023 | Oregon

- Design, build and maintain Spark Streaming pipelines to integrate and warehouse **16+** insurance companies data **ensuring high data integrity** for real-time analytics, reporting and product support.
- Performed **EDA and in-depth analysis** on historical data to identify trends in claims, premiums, and customer behavior over time.
- Define and analyze key performance indicators (KPIs) such as **policy uptake, retention rates, claims ratios, sales** and **customer lifetime that impacts on business success**.
- Developed dashboard and reports to track KPIs in **PowerBI & Tableau** to assess product success.
- Ensured data governance by implementing data quality checks, encryption protocols, and access controls in compliance with industry standards.
- Conducted **A/B tests** for newly released product features, provide insights, and documented results to improve engagement and product performance.
- Collaborated with cross-functional teams in Agile/Scrum environments to deliver data solutions on time and within scope.
- Performed data versioning and documentation of data model, ETL processes and workflow.

Data Analyst (E-Commerce), BitsKraft Pvt. Ltd.

Jan 2020 – Jan 2022 | Kathmandu, Nepal

- Forecasted short and long-term demand at SKU, category, and regional levels using time-series models (**ARIMA**).
- Developed pipeline for data extraction, transformation, **validation, data integrity and quality checks, versioning and warehousing** data documents in NoSQL (MongoDB cluster) using python.
- Performed descriptive analysis on client’s product sales performance, conversion rates, cart abandonment, customer lifetime value (CLV), and average order value (AOV).
- Conducted customer segmentation analysis using **RFM (Recency, Frequency, Monetary)** modeling to identify high-value customers and tailor marketing strategies, resulting in improved retention and targeted campaign.
- Implemented clustering (**K-Means**) to cluster customer, association rule mining (**Apriori**) for market basket analysis, and classification models (**Random Forest, XGBoost**) to classify product performance.
- Worked directly with clients and vendors to identify analytics requirements, document and communicate performance metrics with actionable insights with non-technical stakeholders.

EDUCATION

- **Master’s, Computer Science (GPA: 4.0)** | East Tennessee State University Aug 2023- May 2025 | TN, USA
- **Bachelor’s, Computer Science (GPA: 3.68)** | Tribhuvan University Aug 2016 – May 2020 | Nepal

PROJECTS

- **Supply Chain Analytics:** Using a real-world dataset, the project aims to identify and address key challenges in shipment and inventory management. It analyzes supply chain inefficiencies and delivers interactive dashboards to inform stakeholders and recommend structural improvements. **Tools:** Python, Pandas, NumPy, PowerBI
- **Customer Segmentation Using Clustering:** Applied K-Means and Hierarchical clustering on retail data computing RFM metrics to identify distinct customer groups. Conducted data cleaning, visualization, normalization, and cluster evaluation (silhouette score) to uncover high-value and low-value customer segments. **Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Sentiment360 - Customer Sentiment & Product Insights:** An interactive NLP analytics system that analyzes Amazon product reviews. Performs sentiment classification and LDA-based topic modeling for diagnostic analysis. Identified key pain points across product lines, helping product quality improvements. **Tools:** Python, Pandas, NLP, Streamlit, BERT, PowerBI

PROFESSIONAL DEVELOPMENT

Foundations of Generative AI	Build Data Pipelines with Delta Live Tables
Fundamentals of Accelerated Data Science	Machine Learning with Scikit-Learn.
Data Engineering Foundations	AWS Educate Machine Learning Foundations
Data Science Web App: Streamlit and Python	Getting Started with Data Analytics on AWS
Data Science for Business	Generative AI with AWS

PUBLICATIONS

- **Vision-Based Driver Drowsiness & Distraction Detection Through Behavioral Indicators of Fatigue** (IEEE SoutheastCon 2025, Concord, NC, USA, 2025)
- **Real-Time Driver Drowsiness Detection Using CNN, MediaPipe& ML Classifiers** (IEEE SoutheastCon 2025)
- **AIDE: Leveraging Retrieval-Augmented Generation for Context-Aware Educational Data Retrieval and Dialogue** (IEEE SmartNet 2025, Session: AI/ML for Data Analytics I, Forthcoming (In Press))

PROFESSIONAL ORGANIZATION

IEEE, IEEE Computer Society, Okland East Bay Section	IEEE Member (Present) Member ID: 101189770
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